

# OCEAN taxonomy

Separately coded interface dimensions; labels denote evaluated configurations rather than maturity or performance.

|          |                                      |                                                  |                                                |                                                     |                                              |
|----------|--------------------------------------|--------------------------------------------------|------------------------------------------------|-----------------------------------------------------|----------------------------------------------|
| <b>O</b> | <b>Observation regime</b>            | <b>O0</b><br>Gridded or single source            | <b>O1</b><br>Multimodal observations           | <b>O2</b><br>Irregular / delayed / provenance-aware | <b>O3</b><br>Action-dependent acquisition    |
| <b>C</b> | <b>Constraints and coupling</b>      | <b>C0</b><br>No computational physics insertion  | <b>C1</b><br>Constrained objective / inference | <b>C2</b><br>Physics-structured representation      | <b>C3</b><br>Online solver coupling          |
| <b>E</b> | <b>Evolution</b>                     | <b>E0</b><br>State reconstruction                | <b>E1</b><br>Forecast or emulation             | <b>E2</b><br>Action-conditioned counterfactual      | <b>E3</b><br>Online adaptation with controls |
| <b>A</b> | <b>Agency</b>                        | <b>A0</b><br>Passive analysis                    | <b>A1</b><br>Decision support                  | <b>A2</b><br>Active sensing or control              | <b>A3</b><br>Cooperative multi-agent         |
| <b>N</b> | <b>Nonstationarity and assurance</b> | <b>N0</b><br>Aggregate in-domain evaluation only | <b>N1</b><br>Calibrated uncertainty            | <b>N2</b><br>Declared shift / OOD / fault test      | <b>N3</b><br>Independent assurance envelope  |

D offline/hindcast · S simulation/replay · H hardware/controlled-water · F field. C0/N0 are exclusive baselines; C1-C3 and N1-N3 may coexist.